

## SAFETY DATA SHEET

Creation Date 22-Sep-2009

Revision Date 09-Feb-2024

Revision Number 6

### 1. Identification

**Product Name** Mercury(II) bromide

**Cat No. :** AC448120000; AC448121000

**CAS No** 7789-47-1  
**Synonyms** Mercuric bromide

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

**Company**

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Emergency Telephone Number**For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

**Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                       |            |
|-------------------------------------------------------|------------|
| Acute oral toxicity                                   | Category 2 |
| Acute dermal toxicity                                 | Category 1 |
| Acute Inhalation Toxicity - Dusts and Mists           | Category 2 |
| Specific target organ toxicity - (repeated exposure)  | Category 2 |
| Target Organs - Kidney, Central nervous system (CNS). |            |

**Label Elements****Signal Word**

Danger

**Hazard Statements**

May cause damage to organs through prolonged or repeated exposure

Fatal if swallowed, in contact with skin or if inhaled



### Precautionary Statements

#### Prevention

Wear protective gloves/protective clothing/eye protection/face protection

Use only outdoors or in a well-ventilated area

Wash face, hands and any exposed skin thoroughly after handling

Do not eat, drink or smoke when using this product

Do not get in eyes, on skin, or on clothing

Do not breathe dust/fume/gas/mist/vapors/spray

#### Response

Get medical attention/advice if you feel unwell

#### Inhalation

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

Immediately call a POISON CENTER or doctor/physician

#### Skin

IF ON SKIN: Wash with plenty of soap and water

Immediately call a POISON CENTER or doctor/physician

Remove/Take off immediately all contaminated clothing

Wash contaminated clothing before reuse

#### Ingestion

IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician

Rinse mouth

#### Storage

Store locked up

Store in a well-ventilated place. Keep container tightly closed

#### Disposal

Dispose of contents/container to an approved waste disposal plant

#### Hazards not otherwise classified (HNOC)

Very toxic to aquatic life with long lasting effects

WARNING. Reproductive Harm - <https://www.p65warnings.ca.gov/>.

## 3. Composition/Information on Ingredients

| Component        | CAS No    | Weight % |
|------------------|-----------|----------|
| Mercuric bromide | 7789-47-1 | >95      |

## 4. First-aid measures

|                     |                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>  | Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. |
| <b>Skin Contact</b> | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                       |
| <b>Inhalation</b>   | Remove to fresh air. Immediate medical attention is required. If not breathing, give artificial respiration.                      |
| <b>Ingestion</b>    | Do NOT induce vomiting. Call a physician immediately.                                                                             |

|                                            |                           |
|--------------------------------------------|---------------------------|
| <b>Most important symptoms and effects</b> | No information available. |
| <b>Notes to Physician</b>                  | Treat symptomatically     |

## 5. Fire-fighting measures

|                                         |                                                                              |
|-----------------------------------------|------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray. Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Chemical foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                     |
| <b>Flash Point</b>                      | No information available                                                     |
| <b>Method -</b>                         | No information available                                                     |
| <b>Autoignition Temperature</b>         | No information available                                                     |
| <b>Explosion Limits</b>                 |                                                                              |
| <b>Upper</b>                            | No data available                                                            |
| <b>Lower</b>                            | No data available                                                            |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                     |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                     |

### Specific Hazards Arising from the Chemical

Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Hydrogen halides.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
4

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                             |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required.                                                                                                                                                                 |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. |
| <b>Methods for Containment and Clean Up</b> | Wear self-contained breathing apparatus and protective suit. Sweep up and shovel into suitable containers for disposal. Do not let this chemical enter the environment.                                                                     |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Ensure adequate ventilation. Wear personal protective equipment/face protection. Do not breathe dust. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance. Avoid dust formation. |
| <b>Storage.</b> | Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep locked up. Incompatible Materials. Strong oxidizing agents.                                                                                               |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component        | ACGIH TLV                            | OSHA PEL                                 | NIOSH                                                                                       | Mexico OEL (TWA)             |
|------------------|--------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------|
| Mercuric bromide | TWA: 0.025 mg/m <sup>3</sup><br>Skin | (Vacated) Ceiling: 0.1 mg/m <sup>3</sup> | IDLH: 10 mg/m <sup>3</sup><br>TWA: 0.05 mg/m <sup>3</sup><br>Ceiling: 0.1 mg/m <sup>3</sup> | TWA: 0.025 mg/m <sup>3</sup> |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures** Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

**Personal Protective Equipment**

**Eye/face Protection** Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection** Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:** Particulates filter conforming to EN 143.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                          |
|----------------------------------------|--------------------------|
| Physical State                         | Powder Solid             |
| Appearance                             | White                    |
| Odor                                   | Odorless                 |
| Odor Threshold                         | No information available |
| pH                                     | No information available |
| Melting Point/Range                    | 236 °C / 456.8 °F        |
| Boiling Point/Range                    | 322 °C / 611.6 °F        |
| Flash Point                            | No information available |
| Evaporation Rate                       | Not applicable           |
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | 6.1090                   |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | Br <sub>2</sub> Hg       |
| Molecular Weight                       | 360.4                    |

## 10. Stability and reactivity

**Reactive Hazard** None known, based on information available

**Stability** Light sensitive.

**Conditions to Avoid** Exposure to light. Incompatible products.

|                                         |                                          |
|-----------------------------------------|------------------------------------------|
| <b>Incompatible Materials</b>           | Strong oxidizing agents                  |
| <b>Hazardous Decomposition Products</b> | Hydrogen halides                         |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur. |
| <b>Hazardous Reactions</b>              | None under normal processing.            |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component        | LD50 Oral               | LD50 Dermal | LC50 Inhalation |
|------------------|-------------------------|-------------|-----------------|
| Mercuric bromide | LD50 = 40 mg/kg ( Rat ) | Not listed  | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component        | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|------------------|-----------|------------|------------|------------|------------|------------|
| Mercuric bromide | 7789-47-1 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** Kidney Central nervous system (CNS)

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN1634  
 Proper Shipping Name MERCURY BROMIDES  
 Hazard Class 6.1  
 Packing Group II

#### TDG

UN-No UN1634  
 Proper Shipping Name MERCURY BROMIDES  
 Hazard Class 6.1  
 Packing Group II

#### IATA

UN-No UN1634  
 Proper Shipping Name MERCURY BROMIDES  
 Hazard Class 6.1  
 Packing Group II

#### IMDG/IMO

UN-No UN1634  
 Proper Shipping Name MERCURY BROMIDES  
 Hazard Class 6.1  
 Packing Group II

### 15. Regulatory information

#### United States of America Inventory

| Component        | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|------------------|-----------|------|-----------------------------------------------|-----------------------------|
| Mercuric bromide | 7789-47-1 | X    | ACTIVE                                        | -                           |

#### Legend:

TSCA US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

#### TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)

Not applicable

#### TSCA 12(b) - Notices of Export

Not applicable

#### International Inventories

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component        | CAS No    | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|------------------|-----------|-----|-----|-----------|-------|------|------|------|-------|----------|
| Mercuric bromide | 7789-47-1 | X   | -   | 232-169-3 | X     | -    | X    | X    | X     | KE-23120 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### U.S. Federal Regulations

##### SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

Note that PBT chemicals are not eligible for the de minimis exemption. For these chemicals, supplier notification limits are provided.

> 0 % = no low concentration cut-off set, supplier notification limit applies.

| Component        | CAS No    | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|------------------|-----------|----------|-------------------------------|---------------------------------|
| Mercuric bromide | 7789-47-1 | >95      | > 0 %                         | RT = 10 lb                      |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

| Component        | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|------------------|----------------------------|-----------------------------|------------------------|---------------------------|
| Mercuric bromide | -                          | -                           | X                      | -                         |

**Clean Air Act**

| Component        | HAPS Data | Class 1 Ozone Depleters | Class 2 Ozone Depleters |
|------------------|-----------|-------------------------|-------------------------|
| Mercuric bromide | X         |                         | -                       |

OSHA - Occupational Safety and Health Administration

Not applicable

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65**

This product contains the following Proposition 65 chemicals.

| Component        | CAS No    | California Prop. 65 | Prop 65 NSRL | Category      |
|------------------|-----------|---------------------|--------------|---------------|
| Mercuric bromide | 7789-47-1 | Developmental       | -            | Developmental |

**U.S. State Right-to-Know Regulations**

| Component        | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|------------------|---------------|------------|--------------|----------|--------------|
| Mercuric bromide | -             | X          | X            | X        | -            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant Y

**U.S. Department of Homeland Security**

This product does not contain any DHS chemicals.

**Other International Regulations****Mexico - Grade**

No information available

**Authorisation/Restrictions according to EU REACH**

| Component | CAS No | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High |
|-----------|--------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
|-----------|--------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|

|                  |           |   |                                                                                                                                          | Concern (SVHC) |
|------------------|-----------|---|------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Mercuric bromide | 7789-47-1 | - | Use restricted. See item 18.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | -              |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component        | CAS No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Mercuric bromide | 7789-47-1 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

## Contains component(s) that meet a 'definition' of per &amp; poly fluoroalkyl substance (PFAS)?

Not applicable

## Other International Regulations

| Component        | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Mercuric bromide | 7789-47-1 | Not applicable                                                                            | Not applicable                                                                           | X                          | Annex I - Y29                      |

## 16. Other information

## Prepared By

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## Revision Summary

This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**